Introduction to Artificial Intelligence

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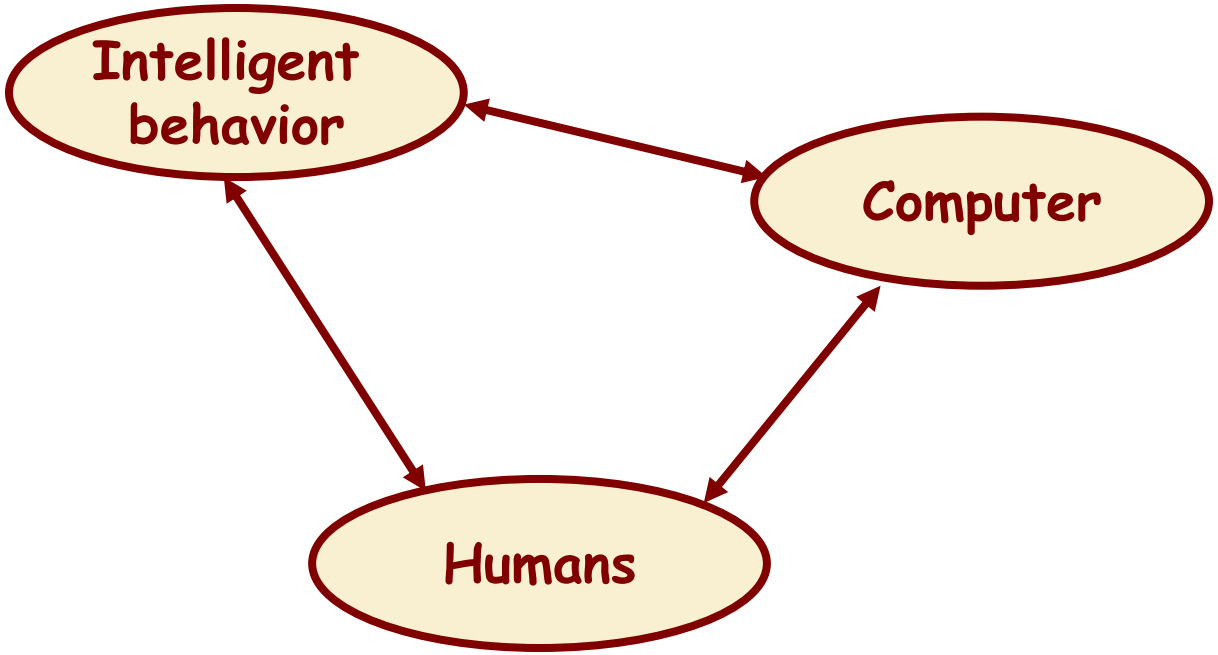
Outline

- What is AI?
- Foundations of AI
- History of Artificial Intelligence
- The AI problem*

What is AI?

an attempt of

- AI is the reproduction of human reasoning and intelligent behavior by computational methods



What is intelligence?

- Behave as intelligently as a human?
- Behave in best possible manner?
- Thinking?
- Acting?

AI Foundations

Mathematics

Psychology

Philosophy

Artificial Intelligence

Computer Science

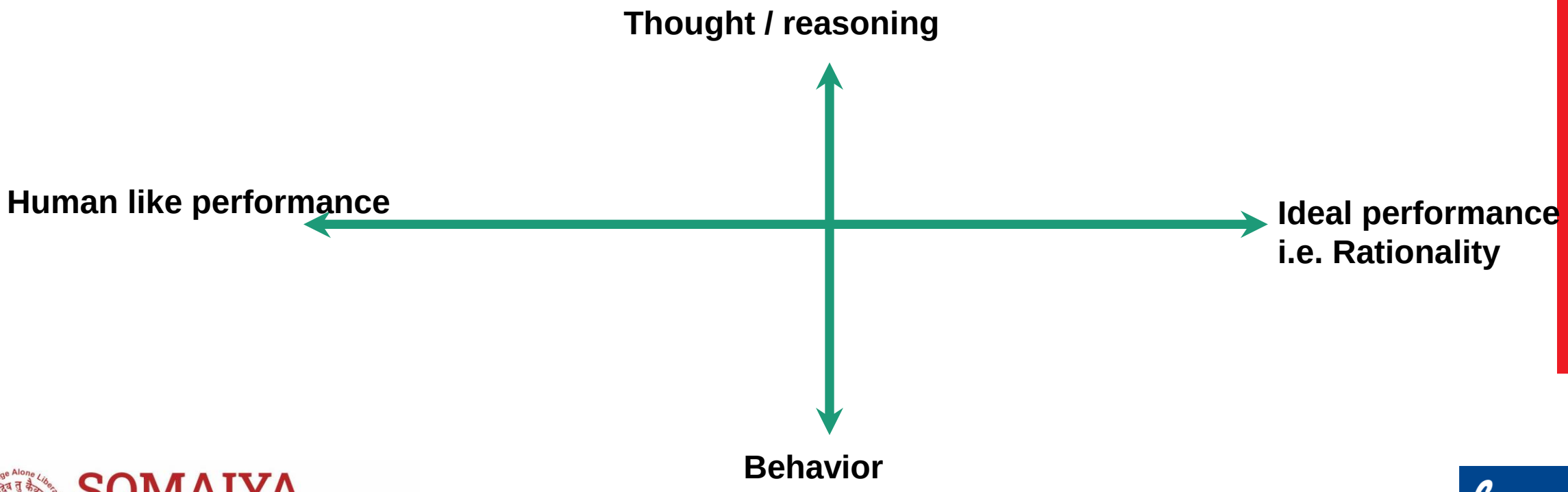
Economics

Biology

Linguistics

AI approaches & Definitions

- Thought process/reasoning Vs behavior
- Human like performance Vs Ideal performance



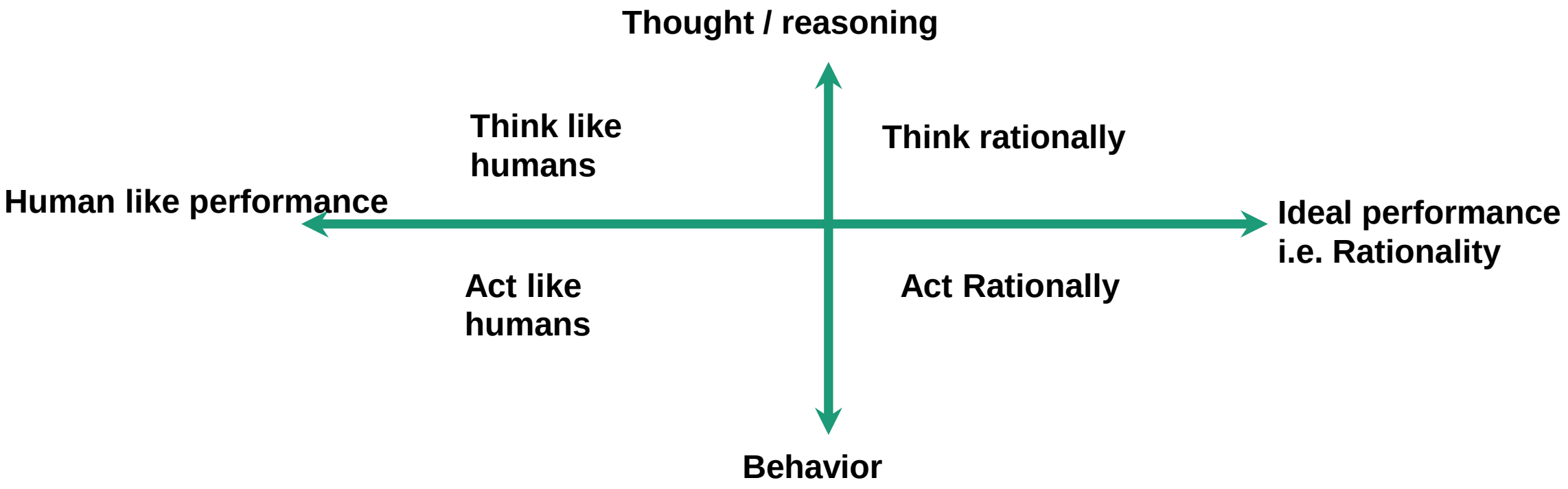
AI approaches & Definitions

1. Systems that think like humans.
2. Systems that think rationally.
3. Systems that act like humans
4. Systems that act rationally

Theories of AI

	Human-like:	Rationally:
Think:	(1) Cognitive science Approach	(2) Laws of thought Approach
Act:	(3) Turing Test Approach	(4) Rational agent Approach

AI approaches & Definitions



Act like humans	Act rationally
Think like humans	Think rationally

- Here, how the computer performs tasks does matter
- The reasoning steps are important
- → Ability to create and manipulate symbolic knowledge (definitions, concepts, theorems, …)
- What is the impact of hardware on low-level reasoning, e.g., to go from signals to symbols?

Act like humans	Act rationally
Think like humans	Think rationally

- The goal of AI is to create computer systems that perform tasks regarded as requiring intelligence when done by humans

- → AI Methodology: Take a task at which people are better, e.g.:
  - Prove a theorem
  - Play chess
  - Plan a surgical operation
  - Diagnose a disease
  - Navigate in a building

  and build a computer system that does it automatically

- But do we want to duplicate human imperfections?

Act like humans	Act rationally
Think like humans	Think rationally

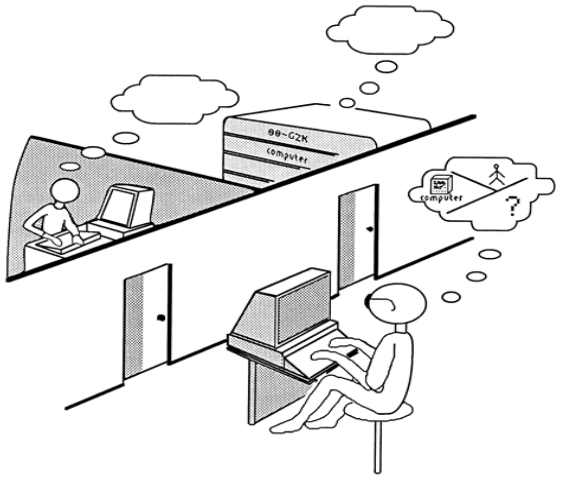
- Now, the goal is to build agents that always make the "best" decision given what is available (knowledge, time, resources)
- "Best" means maximizing the expected value of a utility function
- → Connections to economics and control theory
- What is the impact of self-consciousness, emotions, desires, love for music, fear of dying, etc ... on human intelligence?

Can Machines Act/Think Intelligently?

"If there were machines which bore a resemblance to our bodies and imitated our actions as closely as possible for all practical purposes, we should still have two very certain means of recognizing that they were not real men. The first is that they could never use words, or put together signs, as we do in order to declare our thoughts to others… Secondly, even though some machines might do some things as well as we do them, or perhaps even better, they would inevitably fail in others, which would reveal that they are acting not from understanding, …"

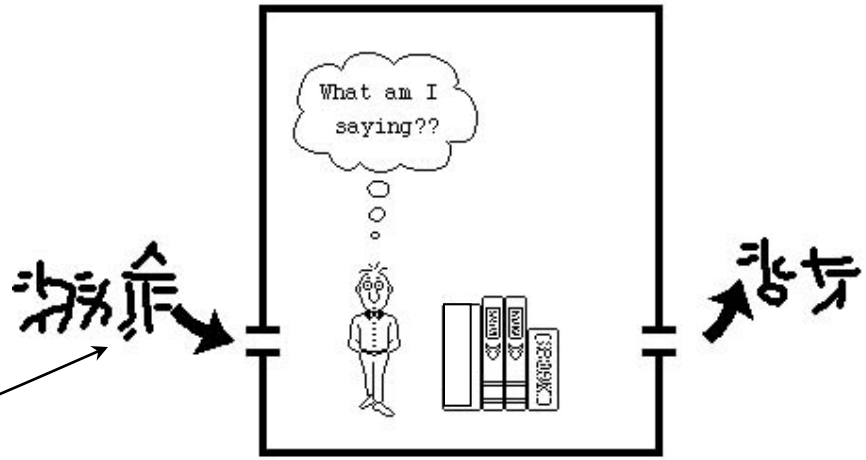
Discourse on the Method, by Descartes (1598-1650)

Can Machines Act/Think Inte



Turing Test:

- http://plato.stanford.edu/entries/turing-test/
- Test proposed by Alan Turing in 1950
- The computer is asked questions by a human interrogator. It passes the test if the interrogator cannot tell whether the responses come from a person
- Required capabilities: natural language processing, knowledge representation, automated reasoning, learning,...
- No physical interaction
- Chinese Room (J. Searle)



Can Machines Act/Think Intelligently?

- Yes, if intelligence is narrowly defined as information processing

  AI has made impressive achievements showing that tasks initially assumed to require intelligence can be automated

  But each success of AI seems to push further the limits of what we consider "intelligence"

Computational Intelligence != Human Intelligence

- In general there are various reasons why trying to mimic humans might **not** be the best approach to AI:
  - Computers and Humans have a very different architecture with quite different abilities.
  - Numerical computations
  - Visual and sensory processing
  - Slow parallel vs. fast serial

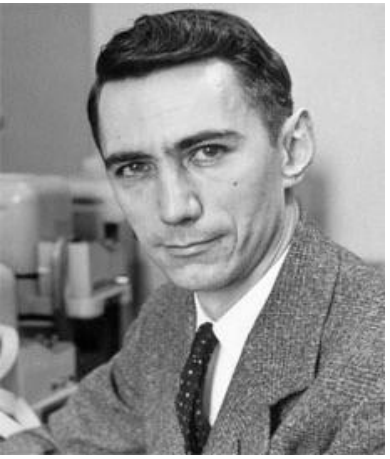
	Computer	Human Brain
Computational Units	8 CPUs, 10^{10} gates	10^{11} neurons
Storage Units	10^{10} bits RAM 10^{13} bits disk	10^{11} neurons 10^{14} synapses
Cycle time	10^{-9} sec	10^{-3} sec
Bandwidth	10^{10} bits/sec	10^{14} bits/sec
Memory updates/sec	10^{10}	10^{14}

Four Categories

Thinking Humanly	**Thinking Rationally**
"The exciting new effort to make computers think . . . *machines with minds*, in the full and literal sense." (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning . . ." (Bellman, 1978)	"The study of mental faculties through the use of computational models." (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act." (Winston, 1992)
Acting Humanly	**Acting Rationally**
"The art of creating machines that perform functions that require intelligence when performed by people." (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better." (Rich and Knight, 1991)	"Computational Intelligence is the study of the design of intelligent agents." (Poole *et al.*, 1998) "AI . . . is concerned with intelligent behavior in artifacts." (Nilsson, 1998)

Figure 1.1 Some definitions of artificial intelligence, organized into four categories.

AI's official birth: Dartmouth, 1956



"An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. ***We think that a significant advance can be made if we work on it together for a summer."***

John McCarthy and Claude Shannon
Dartmouth Workshop Proposal

Shakey the Robot (1966-1972)

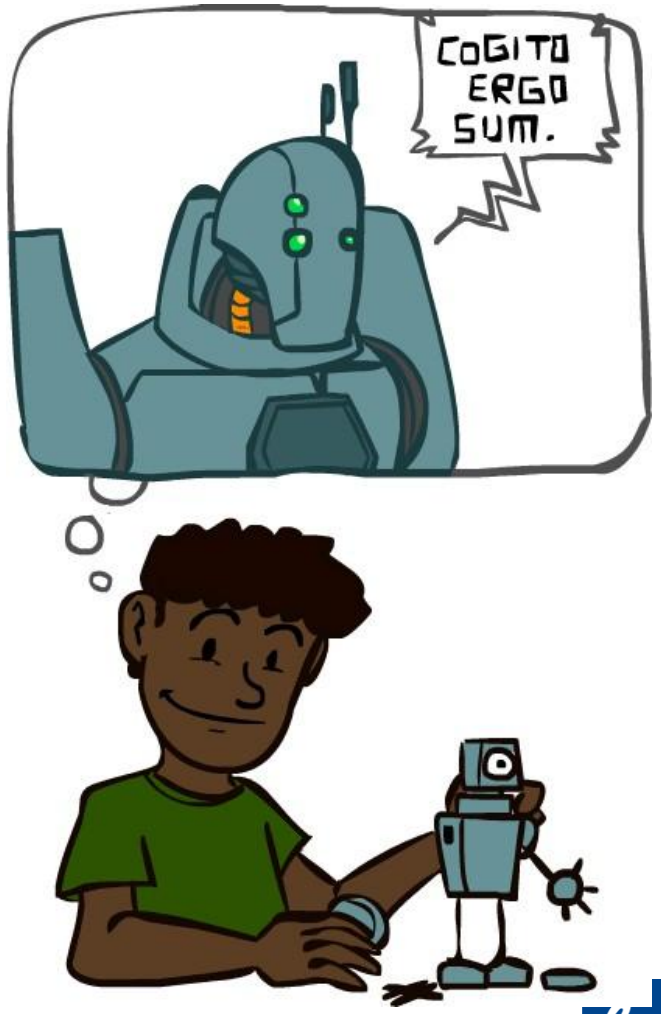
- Short video on Shakey
  - https://www.youtube.com/watch?v=7bsEN8mwUB8

- Long video on Shakey
  - https://www.youtube.com/watch?v=GmU7SimFkpU

Wiki:
https://en.wikipedia.org/wiki/Shakey_the_robot

A (Short) History of AI

- 1940-1950: Early days
  - 1943: McCulloch & Pitts: Boolean circuit model of brain
  - 1950: **Turing's "Computing Machinery and Intelligence"**
- 1950—70: Excitement: Look, Ma, no hands!
  - 1950s: Early AI programs: chess, checkers (RL), theorem proving
  - 1956: **Dartmouth meeting: "Artificial Intelligence" adopted**
  - 1965: Robinson's complete algorithm for logical reasoning
- 1970—90: Knowledge-based approaches
  - 1969—79: Early development of knowledge-based systems
  - 1980—88: Expert systems industry booms
  - 1988—93: Expert systems industry busts: "AI Winter"
- 1990— 2012: Statistical approaches + subfield expertise
  - Resurgence of probability, focus on uncertainty
  - **General increase in technical depth**
  - Agents and learning systems… "AI Spring"?
- 2012—____: Excitement: Look, Ma, no hands again?
  - Big data, big compute, deep learning
  - **AI used in many industries**



A (Short) History of AI

2017: Google AlphaGo beats the world's best Go player, Ke Jie, and AlphaZero learns board games by itself and beats the best programs.

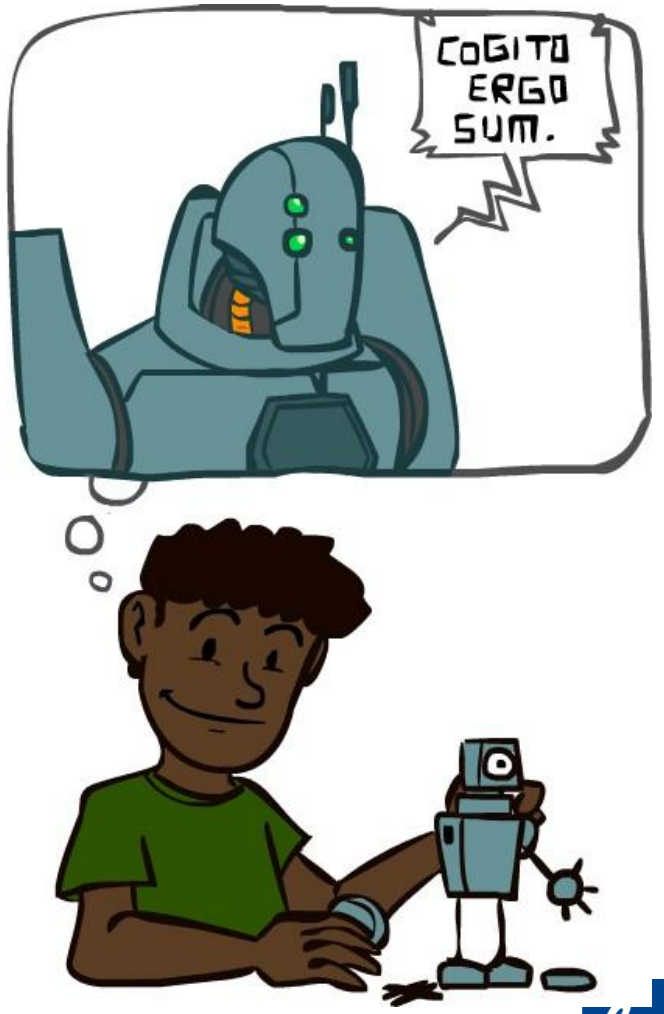
2018: Volvo announces plans to test-drive 100 driverless cars in Gothenburg

2019: GPT-2, developed by OpenAI, showcased the potential of large-scale language models.

2020: OpenAI released GPT-3, significantly advancing natural language processing capabilities.

2021: AI models began to be integrated into various industries, including healthcare, finance, and customer service.

2022: AI ethics and regulations gained prominence, with discussions on bias, transparency, and accountability.

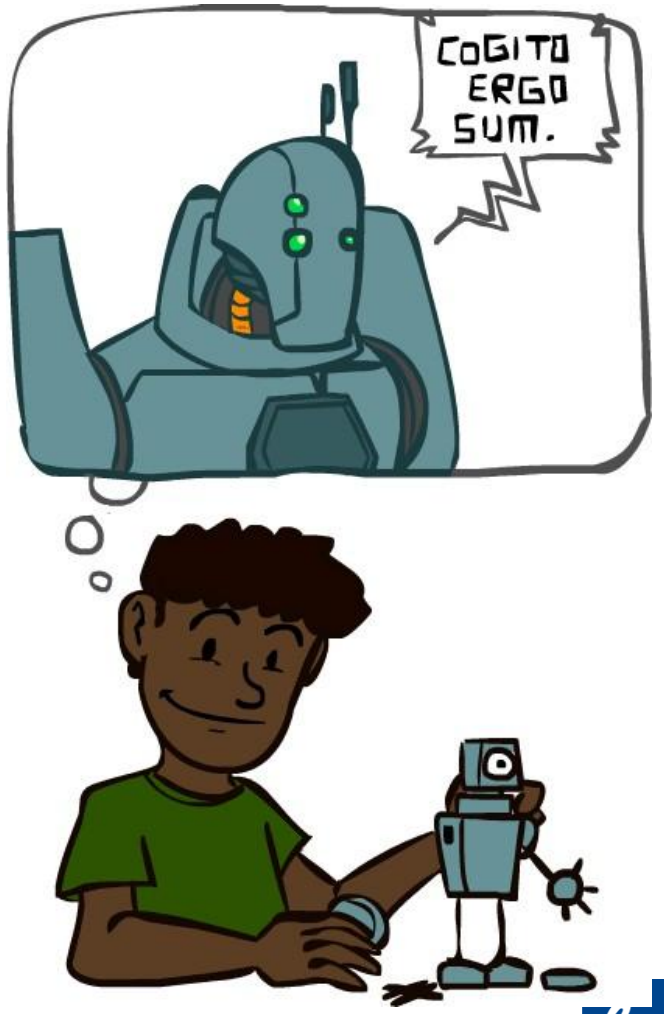


A (Short) History of AI

2023: Advances in AI-driven drug discovery and personalized medicine.

2024:

- OpenAI's Sora: Enhanced text-to-video generation capabilities.
- Google's Gemini: Pushed the boundaries of language model capabilities.
- DeepMind's Veo: Improved multimodal understanding.
- Inflection AI's Pi: Advanced personalized conversational AI.
- RunwayML's Gen-2: Took text-to-video skills to new heights.
- NVIDIA's Market Valuation Milestone: Achieved a $2 trillion market valuation, driven by demand for GPUs powering AI applications.
- Generative AI



The AI Problem

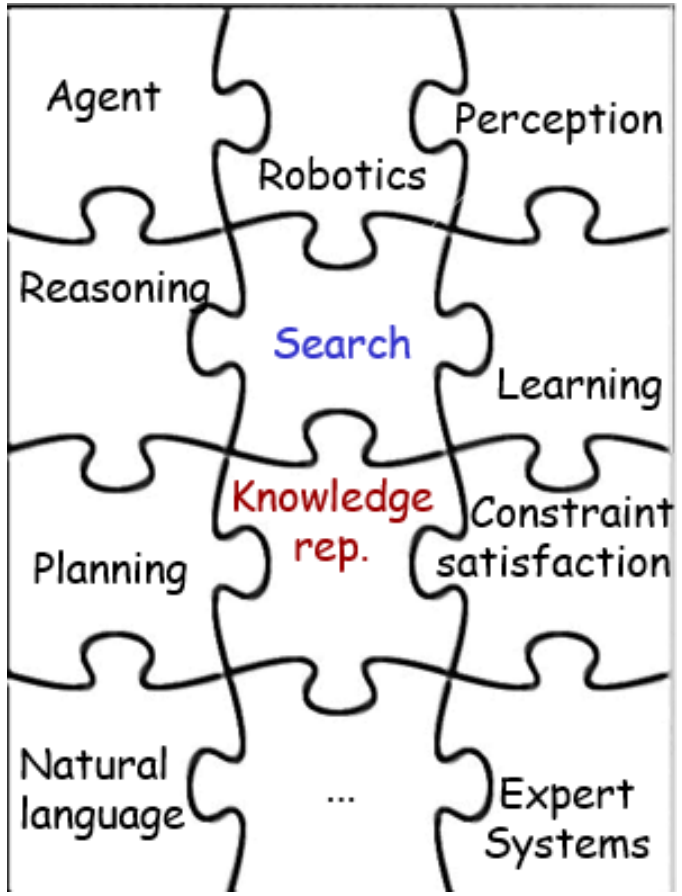
- The main problem is to create intelligent behavior in machines.
- AI must mimic human cognitive abilities: reasoning, learning, problem-solving, perception, and language understanding.

Challenges:

- General vs. Narrow AI
- Uncertainty and Incomplete Information
- Ambiguity in Natural Language

Main Areas of AI

- Knowledge representation (including formal logic)
- Search, especially heuristic search (puzzles, games)
- Planning
- Reasoning under uncertainty, including probabilistic reasoning
- Learning
- Agent architectures
- Robotics and perception
- Natural language processing



Reference

- Artificial Intelligence Modern Approach by Russell and Norvig